

The latent variables could represent additional metabolites in the model with initial conditions that become additional model parameters.

Embedding a mechanistic model into a neural ordinary differential equation produces a physically constrained model that can capture state-dependent shifts in metabolite production and consumption in microbial communities. We used a Bayesian inference approach to enable uncertainty quantification in model parameters and showed how the incorporation of physical constraints increases prediction accuracy and reduces parameter uncertainty. The ability to quantify model uncertainty can be leveraged for Bayesian experimental design, where experimental conditions are selected based on the expected information gain [51]. We envision that this model will play a critical role in accelerating the design-test-learn framework for the discovery of optimized synthetic microbial communities for a broad range of applications spanning precision medicine, sustainable agriculture, environmental cleanup and biomanufacturing and their governing molecular and ecological principles [27].

Methods

The consumer-resource model with growth-associated metabolite production

To generate the simulated data used in Fig. 5, Fig. S1, and Fig. S2, we used a consumer-resource model with growth-associated metabolite production,

$$\begin{aligned}\frac{ds_i(t)}{dt} &= s_i(t) \cdot \left(\sum_{j=1}^{n_m} \epsilon_i C_{ij} m_j(t) - d_i \right) \\ \frac{dm_j(t)}{dt} &= \sum_{i=1}^{n_s} \frac{ds_i(t)}{dt} P_{ij} - m_j \left(\sum_{i=1}^{n_s} s_i(t) C_{ij} \right)\end{aligned}\quad (2)$$

where ϵ_i is the efficiency that species i grows on resources, d_i is the death rate of species i , C_{ij} is the consumption rate of species i on metabolite j , and P_{ij} is the production rate of metabolite j by species i . Species growth efficiencies were randomly sampled as $\epsilon_i \sim \mathcal{U} [.8, .9]$ where $\mathcal{U}$ represents the uniform distribution. Death rates were randomly sampled as $d_i \sim \mathcal{U} [.008, .012]$. Ground truth production rates were randomly generated as $P_{ij} \sim \mathcal{U} [0, 2] \cdot \text{Bernoulli}(.5)$, where $\text{Bernoulli}(.5)$ is a random variable with equal probability of being either zero or one. Each consumption rate was initially sampled as $C_{ij} \sim \mathcal{U} [.5, 1]$. Consumption rates were then set to zero in cases that caused positive feedback loops in metabolite production and consumption. Simulated data was generated by randomly sampling combinations of eight species and six resources where initial species abundances were set to .01 and initial resource concentrations were 1. The data included species abundances and resource concentrations sampled at 10.6, 21.3, and 32 hours, where data was corrupted with zero-mean Gaussian noise with variance $\sigma^2(x) = .001 + .01 \cdot x^2$, where x represents simulated species or metabolite.

The ecological neural ordinary differential equation (eNODE) and neural species mediator (NSM) models

Neural species mediator architecture

We denote the time-dependent vector of species abundances as $\mathbf{s}(t) \in \mathbb{R}_+^{n_s}$, metabolites as $\mathbf{m}(t) \in \mathbb{R}_+^{n_m}$, and additional system inputs as $\mathbf{u} \in \mathbb{R}^{n_u}$. The neural species mediator (NSM) is given by the following set of coupled differential equations,

$$\begin{aligned} \frac{ds_i(t)}{dt} &= s_i(t) \cdot \text{NN}_i(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \phi) \\ \frac{dm_j(t)}{dt} &= \sigma(\text{NN}_j(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \phi)) \cdot \left(\sum_{i=1}^{n_s} \frac{ds_i(t)}{dt} P_{ij} - m_j(t) \left(\sum_{i=1}^{n_s} s_i(t) C_{ij} + b_j \right) \right) \end{aligned} \quad (3)$$

where $P_{ij} \in \mathbb{R}_+$ is the strictly-positive growth-associated rate that species i produces metabolite j , $C_{ij} \in \mathbb{R}_+$ is the strictly-positive rate that species i consumes metabolite j , and $b_j \in \mathbb{R}_+$ is the constant degradation rate of metabolite j . We use the rectified linear unit function, $a^+ = \max(0, a)$, to model metabolite production as proportional to species growth. The structure of the mechanistic component can be modified as follows to account for biomass associated metabolite production where metabolites can be produced during stationary and death phase,

$$\begin{aligned} \frac{ds_i(t)}{dt} &= s_i(t) \cdot \text{NN}_i(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \phi) \\ \frac{dm_j(t)}{dt} &= \sigma(\text{NN}_j(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \phi)) \cdot \left(\sum_{i=1}^{n_s} s_i(t) P_{ij} - m_j(t) \left(\sum_{i=1}^{n_s} s_i(t) C_{ij} + b_j \right) \right). \end{aligned} \quad (4)$$

Neural network architecture

The neural network in the eNODE and NSM (Eq. 1) uses a feedforward architecture to predict the rate of change of species and metabolites,

$$\text{NN}(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \phi) := \mathbf{W}_2 \cdot \tanh \left(\mathbf{W}_1 \cdot \begin{bmatrix} \mathbf{s}(t) \\ \mathbf{m}(t) \\ \mathbf{u} \end{bmatrix} + \mathbf{b}_1 \right) + \mathbf{b}_2 \quad (5)$$

The set of weights and biases in the neural network is denoted as $\phi = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2\}$.

Constraining mechanistic model parameters

The parameters in the consumer-resource model are all strictly non-negative and represent rates of metabolite production, consumption, and degradation. To ensure non-negativity of rate parameters, we apply the SOFTPLUS function with base ten, $\sigma_{10}(a) =$

532 $\log_{10}(1 + 10^a)$, to each parameter. We denote the unconstrained parameter using a hat,
 533 e.g. $C_{ij} = \sigma_{10}(\hat{C}_{ij})$. The full set of parameters in the NSM is denoted as $\theta = \{\hat{P}, \hat{C}, \hat{d}\} \cup \phi$.

534 Parameter estimation

535 General description of data structure and model predictions

536 The central modeling goal in this study is to predict the distribution of experimental depen-
 537 dent variables given all independent variables. In this case, dependent variables include
 538 species abundances and metabolite concentrations and independent variables include
 539 initial conditions of species and metabolites, media conditions such as the presence or
 540 absence of resources, and the measurement time. We denote an experimental condi-
 541 tion as the set of independent variables, $x = \{\mathbf{s}(0), \mathbf{m}(0), \mathbf{u}, \tau\}$. We denote the vector of
 542 measured dependent variables as

$$\mathbf{y}(x) = \begin{bmatrix} \mathbf{s}(\tau) \\ \mathbf{m}(\tau) \end{bmatrix}. \quad (6)$$

543 We denote model predictions of the dependent variables as $\hat{\mathbf{y}}(x)$,

$$\begin{aligned} \hat{\mathbf{y}}(x) &= \begin{bmatrix} \hat{\mathbf{s}}(\tau) \\ \hat{\mathbf{m}}(\tau) \end{bmatrix} \\ &= \int_0^\tau \mathbf{g}(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \theta) dt + \begin{bmatrix} \mathbf{s}(0) \\ \mathbf{m}(0) \end{bmatrix}, \end{aligned} \quad (7)$$

544 where $\mathbf{g}(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \theta)$ is a model such as the NSM, eNODE, gLV, or CR, that predicts
 545 rate of change of species and metabolites. A data set is composed of measured species
 546 abundances and metabolite concentrations in n unique experimental conditions, which
 547 we denote as $\mathcal{D} = \{\mathbf{y}(x_1), \dots, \mathbf{y}(x_n)\}$.

Maximum likelihood

Prediction of species abundances and metabolite concentrations is a regression problem that involves optimizing model parameters such as species growth rates, metabolite consumption rates, and neural network weights and biases, such that the error between model predictions and measured values is minimized. Experimental measurements of species and metabolites are inherently noisy, meaning that repeated measurements of the same variable under the same experimental conditions will result in a distribution of values. In practice, this distribution is approximately Gaussian with a standard deviation that scales linearly with the magnitude of the measured variable. We therefore model the conditional distribution of an experimental measurement in a particular experimental condition as a Gaussian with a mean and variance predicted by the model,

$$p(\mathbf{y}|x_i, \theta, \nu, \eta) = \prod_{j=1}^{n_s+n_m} \mathcal{N}(y_j(x_i)|\hat{y}_j(x_i, \theta), \nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta)^2). \quad (8)$$

Given this model for the distribution of a measured variable, the likelihood of a set of independent measurements is

$$p(\mathcal{D}|\mathbf{x}, \theta, \nu, \eta) = \prod_{i=1}^n \prod_{j=1}^{n_s+n_m} \mathcal{N}(y_j(x_i)|\hat{y}_j(x_i, \theta), \nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta)^2). \quad (9)$$

Maximizing the likelihood of the data with respect to parameters is equivalent to maximizing the log likelihood and gives the maximum likelihood estimate (MLE),

$$\theta_{MLE} = \arg \max_{\theta} \sum_{i=1}^n \sum_{j=1}^{n_s+n_m} \left(-\frac{1}{2} \cdot \frac{(y_j(x_i) - \hat{y}_j(x_i, \theta))^2}{\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta)^2} - \frac{1}{2} \cdot \log(\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta)^2) \right). \quad (10)$$

In the case where measurement noise is assumed constant, the maximum likelihood estimate is equivalent to minimizing the sum of squared error between model predictions and measured values,

$$\theta_{MLE} = \arg \min_{\theta} \sum_{i=1}^n \sum_{j=1}^{n_s+n_m} (y_j(x_i) - \hat{y}_j(x_i, \theta))^2. \quad (11)$$

Maximum A Posteriori

Maximum likelihood parameter estimation can suffer poor generalization due to the model fitting to measurement noise. To mitigate this, a prior on parameters centered at zero promotes simplicity in the model. Furthermore, using an independent Gaussian distribution centered at zero for each parameter prior promotes sparsity by driving the posterior distribution of unnecessary parameters to the prior mean [4]. We use a Gaussian prior centered at zero with a variance for each parameter, $p(\theta) = \prod_{i=1}^{n_{\theta}} \mathcal{N}(\theta_i|0, \alpha_i^{-1})$. Using Bayes' theorem gives the posterior probability of parameters,

$$p(\theta|\mathcal{D}, \mathbf{x}, \alpha, \nu, \eta) \propto p(\mathcal{D}|\mathbf{x}, \theta, \nu, \eta) \cdot p(\theta|\alpha). \quad (12)$$

Maximizing the posterior probability gives the maximum a posteriori (MAP) estimate,

$$\begin{aligned} \theta_{MAP} = \arg \max_{\theta} \sum_{i=1}^n \sum_{j=1}^{n_s+n_m} \left(-\frac{1}{2} \cdot \frac{(y_j(x_i) - \hat{y}_j(x_i, \theta))^2}{\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta)^2} - \frac{1}{2} \cdot \log(\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta)^2) \right) \\ - \sum_{k=1}^{n_{\theta}} \frac{1}{2} \alpha_k \theta_k^2. \end{aligned} \quad (13)$$

Variational inference

The goal of Bayesian inference is to determine the conditional probability distribution of model parameters given a data set, which is termed the posterior parameter distribution. Variational inference is a method that casts estimating the posterior parameter distribution as an optimization problem where the objective is to minimize the Kullback-Leibler divergence between an approximate parametric probability density function and the true parameter posterior. A popular choice for a probability density function is an independent

582 Gaussian for each parameter due to its analytical tractability and computational scalabil-
 583 ity, denoted here as $q(\theta|\mu, \sigma) = \prod_{i=1}^{n_\theta} \mathcal{N}(\theta_i|\mu_i, \sigma_i^2)$. To enable a more compact notation we
 584 denote the posterior as $p(\theta|\mathcal{D}, \alpha, \nu, \eta)$, omitting the dependence on the set of experimen-
 585 tal conditions, $\mathbf{x}$. The Kullback-Leibler divergence between the approximate posterior and
 586 the true posterior is

$$\begin{aligned} \text{KL}[q(\theta|\mu, \sigma)||p(\theta|\mathcal{D}, \alpha, \nu, \eta)] &= \int_{\theta} q(\theta|\mu, \sigma) \log \frac{p(\theta|\mathcal{D}, \alpha, \nu, \eta)}{q(\theta|\mu, \sigma)} d\theta \\ &= \int_{\theta} q(\theta|\mu, \sigma) \log p(\theta|\mathcal{D}, \alpha, \nu, \eta) d\theta - \int_{\theta} q(\theta|\mu, \sigma) \log q(\theta|\mu, \sigma) d\theta \\ &= \int_{\theta} q(\theta|\mu, \sigma) \log p(\mathcal{D}|\theta, \nu, \eta) d\theta + \int_{\theta} q(\theta|\mu, \sigma) \log p(\theta|\alpha) d\theta \\ &\quad - \int_{\theta} q(\theta|\mu, \sigma) \log p(\mathcal{D}|\alpha, \nu, \eta) d\theta - \int_{\theta} q(\theta|\mu, \sigma) \log q(\theta|\mu, \sigma) d\theta. \end{aligned} \quad (14)$$

587 Keeping only the terms that depend on the variational parameters μ and σ , we define

$$\mathcal{L}(\mu, \sigma) = \int_{\theta} q(\theta|\mu, \sigma) \log p(\mathcal{D}|\theta, \nu, \eta) d\theta + \int_{\theta} q(\theta|\mu, \sigma) \log p(\theta|\alpha) d\theta - \int_{\theta} q(\theta|\mu, \sigma) \log q(\theta|\mu, \sigma) d\theta. \quad (15)$$

588 The function $\mathcal{L}(\mu, \sigma)$ is called the evidence lower bound (ELBO). Although the expectation
 589 of the log likelihood is analytically intractable, the ELBO can be estimated using a Monte
 590 Carlo approximation,

$$\begin{aligned} \mathcal{L}(\mu, \sigma) &\approx \frac{1}{m} \sum_{\ell=1}^m \sum_{i=1}^n \sum_{j=1}^{n_s+n_m} \left(-\frac{1}{2} \cdot \frac{(y_j(x_i) - \hat{y}_j(x_i, \theta^{(\ell)}))^2}{\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta^{(\ell)})^2} - \frac{1}{2} \cdot \log(\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta^{(\ell)})^2) \right) \\ &\quad - \frac{1}{m} \sum_{\ell=1}^m \sum_{k=1}^{n_\theta} \frac{1}{2} \alpha_k \theta_k^{(\ell)2} + \sum_{i=1}^{n_\theta} \log \sigma_i. \end{aligned} \quad (16)$$

where $\theta^{(\ell)} \sim q(\theta|\mu, \sigma)$. Optimization of the ELBO with respect to μ and σ is performed using the ADAM algorithm with the default parameters of $\alpha = .001$, $\beta_1 = .9$, and $\beta_2 = .999$ [25]. The slope of a linear fit to the ELBO over the previous 100 training epochs is used to evaluate convergence. The algorithm terminates when the slope of the ELBO is less than .001 for five consecutive evaluations.

Expectation Maximization to optimize hyperparameters

Model hyperparameters include the precision of the parameter prior, α , and the measurement noise variance parameters ν and η . The expectation maximization (EM) algorithm iterates between inference of the parameter posterior distribution and then maximization of the expected joint distribution of the data and model parameters with respect to model hyperparameters [4]. Because inference of the posterior depends on the hyperparameters, the posterior distribution needs to be updated given the optimized hyperparameters. The process of iterating between updating the posterior and updating hyperparameters continues until convergence of the ELBO. The expectation of the joint distribution of the data and model parameters is

$$\begin{aligned} \mathcal{Q}(\alpha, \nu, \eta) &= \int_{\theta} q(\theta|\mu, \sigma) \log p(\mathcal{D}, \theta|\alpha, \nu, \eta) d\theta \\ &= \int_{\theta} q(\theta|\mu, \sigma) \log p(\mathcal{D}|\theta, \nu, \eta) d\theta + \int_{\theta} q(\theta|\mu, \sigma) \log p(\theta|\alpha) d\theta \\ &\approx \frac{1}{m} \sum_{\ell=1}^m \sum_{i=1}^n \sum_{j=1}^{n_s+n_m} -\frac{1}{2} \cdot \frac{(y_j(x_i) - \hat{y}_j(x_i, \theta^{(\ell)}))^2}{\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta^{(\ell)})^2} - \frac{1}{2} \cdot \log(\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta^{(\ell)})^2) \\ &\quad - \frac{1}{m} \sum_{\ell=1}^m \sum_{k=1}^{n_{\theta}} \frac{1}{2} \left(\alpha_k (\theta_k^{(\ell)})^2 - \log \alpha_k \right) \end{aligned} \quad (17)$$

where $\theta^j \sim \mathcal{N}(\theta|\mu, \sigma)$. Maximization of $\mathcal{Q}(\alpha, \nu, \eta)$ with respect to α gives the update equation,

$$\alpha_k = \frac{1}{\frac{1}{m} \sum_{\ell=1}^m (\theta_k^{(\ell)})^2}. \quad (18)$$

608 $\mathcal{Q}(\alpha, \nu, \eta)$ is maximized with respect to ν and η when

$$\nu_j^2 + \eta_j^2 \cdot \hat{y}_j(x_i, \theta^{(\ell)})^2 = (y_j(x_i) - \hat{y}_j(x_i, \theta^{(\ell)}))^2 \quad (19)$$

609 for each $j = 1, \dots, n_s + n_m$ for all $\ell = 1, \dots, m$ and $i = 1, \dots, n$.

610 Models for comparison with the NSM

611 Generalized Lotka-Volterra (gLTV)

$$\frac{ds_i(t)}{dt} = s_i(t) \cdot \left(r_i + \sum_{j=1}^{n_s} a_{ij} s_j(t) \right) \quad (20)$$

612 Consumer-resource (CR)

$$\begin{aligned} \frac{ds_i(t)}{dt} &= s_i(t) \cdot \left(\sum_{j=1}^{n_m} C_{ij} m_j(t) - d_i \right) \\ \frac{dm_j(t)}{dt} &= \sum_{i=1}^{n_s} \frac{ds_i(t)}{dt} P_{ij} - m_j \left(\sum_{i=1}^{n_s} s_i(t) C_{ij} + b_j \right) \end{aligned} \quad (21)$$

613 Universal ordinary differential equation (UODE)

$$\begin{aligned} \frac{ds_i(t)}{dt} &= s_i(t) \cdot \text{NN}_i(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \phi) \\ \frac{dm_j(t)}{dt} &= \text{NN}_j(\mathbf{s}(t), \mathbf{m}(t), \mathbf{u}, \phi) + \left(\sum_{i=1}^{n_s} \frac{ds_i(t)}{dt} P_{ij} - m_j(t) \left(\sum_{i=1}^{n_s} s_i(t) C_{ij} + b_j \right) \right) \end{aligned} \quad (22)$$

614 Strain culturing and hydrogen sulfide quantification

615 Strain maintenance and culturing of *Faecalibacterium prausnitzii* (FP) DSMZ 17677 and
616 *Desulfovibrio piger* (DP) ATCC 29098 were performed according to Clark 2021 [12]. Side-

by-side monoculture and coculture growth kinetics were measured in response to varying key nutrients (glucose, lactate, and sulfate) according to 3-factor 3-level full factorial experimental design. The variable nutrients were assembled according to the experimental design using a Tecan Evo Liquid handling robot (see Supplemental data file S?? for base medium composition and variable component design levels). The media used were based on that developed in Clark 2021 [12]. Replicate plates copies corresponding to each sample timepoint were prepared, inoculated, and cultured in parallel, with one plate harvested at each timepoint. This was done so as to avoid depletion of culture volume across multiple measurements from small volume cultures and to avoid disturbing apparent three-dimensional structure of the cultures (which we expected may be contribute to interspecies-interactions). Identically prepared cultures were each harvested at 7, 10, 19, and 37 hour timepoints, each containing n=3 biological replicates of each media-species or media-coculture combination. Measurements of all biological replicates are included in Supplementary Data 1. Inoculation density for monoculture conditions was .01 OD600 and coculture experiments was .005 OD600 per species (.01 OD600 total). Hydrogen sulfide production was measured at each timepoint using a 96-well plate-based adaption of the cline assay according to Hromada 2023 [21]. Coculture relative abundance was measured by 16S rRNA sequencing on an illumine MiSeq according to detailed methods in Clark 2021 [12]. Total growth was measured by optical density at 600nm, diluting and back calculating if greater than the linear range of the instrument (1.0 OD600).

Correlated coefficients z-test

We use the correlated coefficients z-test to test whether there is a significant difference in correlations between measured and predicted values of two competing models [36]. The test yields a z-score testing the null hypothesis of equal correlation coefficients, accounting for correlations between the two model predictions.